

A Drug-Drug Interaction Prediction Method Based on Atomic 3D Position Encoding and Elastic Message Passing Graph Neural Network

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Abstract—Drug-drug interaction (DDI) refers to the inhibitory or enhancing effects between different drugs. Existing DDI prediction methods primarily use graph neural networks (GNNs) to directly represent drug molecular features. However, they often ignore the 3D structures of different atoms within drug molecules and the impact of noise in GNNs on DDI prediction. Consequently, the accuracy of GNN-based DDI prediction remains unsatisfactory. To address these limitations, this study proposes a DDI prediction method based on atomic 3D position encoding and an elastic message passing graph neural network (A3DPE-EMPGNN). Firstly, we construct an atomic feature network based on an attention mechanism and a message passing neural network. This network leverages 3D position encoding based on the molecular centroid to learn the features of different atoms and their associated chemical bonds, thereby constructing a graph-based molecular representation. Secondly, we design a molecular feature network that incorporates an attention mechanism, utilizing multi-head attention to capture interaction information between different drug molecules. Thirdly, we employ an adversarial attack detection and defense strategy, integrating supervised and contrastive loss learning to optimize the model and enhance its robustness while performing DDI prediction. Lastly, we evaluate the effectiveness of A3DPE-EMPGNN on two real-world datasets. Experimental results clearly demonstrate that our method achieves over 98% accuracy across ACC, AUC, AP, and F1-score, outperforming state-of-the-art GNN-based models.

Index Terms—3D position encoding, drug-drug interaction prediction, elastic message passing graph neural network, multi-layer attention mechanism, network attack detection and defense strategy.

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I. INTRODUCTION

DRUG-DRUG interaction refers to the phenomenon where the performance of certain drugs is affected by the presence of other drugs when two or more drugs are combined, resulting in either synergistic or antagonistic effects [1], [2]. DDIs can lead to side effects that may reduce the efficacy of drugs, induce adverse reactions, or even negatively impact the health of patients. Traditional experimental methods for DDI prediction are costly and time-consuming [3], [4]. In recent years, computational methods aided by computer technologies have gained increasing application [5], [6]. By leveraging computer-assisted computational methods to learn from existing drug interaction data and build predictive models, the efficiency of DDI prediction tasks can be greatly improved [7]. Therefore, computer-aided DDI prediction methods have significant theoretical value, as they can deepen the understanding of drug interaction mechanisms and promote the development of drug science theories. At the same time, they also have great practical value, as they can improve the efficiency and accuracy of DDI prediction, reduce the cost and risk of drug development, and promote the advancement of personalized medicine.

The DDI prediction task primarily involves quantifying drug molecular formulas, extracting their features, and selecting an appropriate model to predict DDIs [8], [9]. Unlike general structured data, drug molecules possess definite molecular structures and biochemical properties. Therefore, the key to solving the DDI prediction problem lies in learning the structure and feature information of drug molecules and building a predictive model. Currently, there have been numerous research achievements related to DDI prediction tasks, and the methods involved can be broadly categorized into traditional machine learning-based approaches and deep learning-based approaches [10], [11], [12].

With the development of deep learning technology, graph-based neural network methods have been proposed by researchers and successfully applied to DDI prediction. These methods utilize biochemical feature information of drugs, including drug targets, enzymes, transporters, and proteins, to construct a graph structure for drug molecules by treating different drug molecules as nodes and their interactions as edges, thereby enabling DDI prediction [13], [14], [15]. By fully utilizing the associated biochemical feature information of drugs, these methods can improve the accuracy of DDI prediction. However, they also have the following limitations: 1. Limited

representation of 3D structural information: In existing DDI prediction methods, SMILES can only describe the connectivity between atoms through a linear sequence, but it cannot represent the relative positions, angles, and spatial configurations of atoms in three-dimensional space. The lack of 3D structural information prevents the prediction models from fully understanding the physicochemical properties of drug molecules and their interactions with target proteins or other drug molecules. This is particularly crucial in DDI prediction, where spatial contact and interaction patterns between drugs, such as contact areas and angles, are key factors. When modeling is based solely on the one-dimensional linear SMILES method, existing approaches fail to accurately capture the true spatial relationships and interaction patterns between drug molecules, which consequently affects the accuracy of DDI predictions. 2. Inadequate integration of atomic and molecular-level features: Due to the involvement of multiple drug molecules and atomic information in DDI prediction tasks, existing methods have failed to deeply explore the long-range dependencies between atoms within drug molecules and the specific interaction patterns between molecules. Typically, GNN models are used to fuse feature information for each atom within a drug molecule, which usually includes atomic type, electronegativity, bond type, and other physicochemical properties, as well as the neighboring atomic environment. At the molecular level, features such as molecular weight, molecular rings, and functional group information are fused through global graph embeddings derived from molecular substructures. However, they fail to fully integrate long-range atomic relationships, inter-molecular interactions, 3D structural and dynamic feature information. This limitation leads to insufficiently comprehensive fusion of feature information, which consequently affects the accuracy of DDI predictions. 3. Challenges in maintaining data quality and model robustness: When GNNs are used for drug atom and molecular representations, they often ignore the potential effects of noise, data contamination, and network attacks. In practical settings, experimental data or structural information may contain inaccuracies, and models can be vulnerable to adversarial interference, which could degrade the quality of data inputs. The presence of noisy or inconsistent data can lead to feature contamination, misrepresentation of the molecular structure, and interference within the graph structure. This undermines the model's ability to generalize, increasing the risk of overfitting to the training data and reducing its robustness. As a result, the model may struggle to make accurate and reliable predictions in real-world applications, especially when dealing with complex or noisy datasets.

To address the above issues, we propose a DDI prediction method based on multi-layer attention mechanism and elastic message passing graph neural network. To fully explore the molecular structure information of drugs, we study the DDI prediction method from two levels: within drug molecules (among different atoms) and between different drug molecules. By utilizing a message passing neural network based on the attention mechanism, we learn the graph structural features of drug molecules, including different atoms and chemical bonds. Additionally, we extract feature information on the relationships between different drug molecules using a neural network with a

multi-head attention mechanism. This enables the extraction of drug molecular features from two distinct levels—within and between drug molecules—thereby completing the DDI prediction task.

The main contributions of our work can be summarized as follows:

- We propose a 3D position encoding strategy based on the molecular centroid, using a multi-layer attention mechanism and elastic message passing graph neural network to learn different interaction information between atoms, molecules, and their associated chemical bonds.
- By considering the diverse interaction relationships between molecules, we propose a molecular feature network based on an attention mechanism and dense connection structure, enabling drug learning and DDI prediction at both the atomic and molecular levels.
- We introduce a new structured learning method for an elastic graph neural network to enhance the robustness of the graph neural network. This method combines an elastic message passing scheme, a threshold-based edge deletion strategy, and a structured learning process, effectively detecting and defending against potential network attacks.

II. RELATED WORK

Computer-aided DDI prediction methods can be broadly categorized into traditional machine learning-based approaches and deep learning-based approaches. Traditional machine learning methods mainly fall into three categories: methods based on traditional classifiers, regression-based methods, and matrix factorization-based methods [16], [17], [18]. In the traditional classifier-based and regression-based methods, similarity measurement methods (such as inner product-based or cosine similarity measures) are typically used to assess the similarity between two drugs, and different classification or regression algorithms are applied to predict the probability of interactions between different drugs [19], [20]. In matrix factorization-based methods, the DDI prediction task can be modeled as a matrix completion task, where the goal is to recover a complete matrix from an incomplete one, predicting unobserved interactions [16]. Traditional machine learning methods have limitations in extracting these deep features because they often rely on simplified feature representations and superficial information, making it difficult to delve into the complex atomic features, spatial structural features, chemical bond features, and interaction patterns between drug molecules. The absence of these deep features restricts the predictive accuracy of traditional methods on large-scale or imbalanced datasets. This results in inadequate prediction accuracy, especially on imbalanced and large-scale datasets [21]. Moreover, these methods often directly use drug molecular feature information, neglecting the atomic-level features within drug molecules, which also affects the performance of DDI prediction.

Unlike traditional machine learning-based methods, deep learning-based DDI prediction methods can extract deeper features and often perform better in predicting potential DDIs in

real-world applications [22], [23], [24]. For instance, Ryu et al. [25] proposed a deep learning-based DDI prediction model that predicts DDIs by learning structural similarity information between different drugs. Deng et al. [26] combined drug structure, gene ontology, and target gene similarity information with a deep neural network to extract drug molecular features for DDI prediction. Lee et al. [27] used different similarity information for each drug to train a model and employed an autoencoder and deep feedforward network to predict DDIs. While these methods leverage the deep feature extraction capabilities of deep learning, they still fail to capture the graph-structured nature of drug data, resulting in suboptimal performance in practical applications.

In recent years, graph-structured deep learning methods have been proposed and successfully applied to DDI prediction [28], [29], [30]. These methods construct a graph network for drug molecules by treating different drug molecules as nodes and interaction relationships as edges, thereby enabling DDI prediction. For example, Ziaee et al. [31] proposed a predictive model by integrating node2vec, word2vec, indications, side effects, drug fingerprints, and drug targets through a graphical autoencoder model. This model uses GNN to prioritize potentially novel and effective combination therapies. Liu et al. [32] represented drugs and proteins as nodes in a heterogeneous graph, where different types of edges not only represent DDI, but also Drug-Protein Interactions (DPI) and Protein-Protein Interactions (PPI). In addition, to analyze the chemical structure of drugs, they used a pre-trained model called ChemBERTa, which utilizes the RDKit tool to determine the similarity between drug structures based on SMILES strings. Lin et al. [33] designed a multi-layer adaptive soft-mask graph neural network to extract substructures from molecular graphs and employed an attention mechanism to mine substructure feature information, updating latent features. During this process, to optimize the final feature representation, they decomposed drug interactions into pairwise interaction relationships between core substructures of drugs.

Although these methods take into account the graph structure information of drug molecules, they also have drawbacks. The aforementioned method extracts features at the molecular level of drugs while ignoring feature learning at the atomic level within drug molecules and their 3D structures. However, the 3D structure of a drug molecule plays a crucial role in deeply understanding its physicochemical properties, predicting drug-target interactions, and revealing specific interaction patterns between drugs. The 3D structure not only determines the spatial conformation of the drug but also affects its binding affinity, selectivity, and potential synergistic or antagonistic effects with other drugs. In many existing DDI prediction methods, the structure of drug molecules is typically encoded using the one-dimensional linear representation of SMILES. However, this representation fails to effectively capture the spatial relationships between atoms within the drug molecule. The atoms within a drug molecule may exhibit different behaviors in chemical reactions due to variations in their 3D structure, thereby influencing their interactions with other drugs or targets. Moreover, these methods do not consider the impact of noise interference on graph neural networks, leading to poor model robustness. To address these issues, we propose a drug interaction prediction method based

on multi-layer attention mechanism and elastic message passing graph neural network.

III. MODEL

The DDI prediction task we are studying is to predict the interaction relationship between any two drug molecules in a given set of drug molecules. We constructed a drug interaction prediction model using a multi-layer attention mechanism and an elastic message passing graph neural network. First, we extract the graph structure information of drug molecules from the 3D structure of the drug molecules, including features associated with atoms and chemical bonds [34]. Then, at the atomic feature level, we utilize an elastic message passing graph neural network combined with the attention mechanism of the Transformer and our proposed 3D position encoding method based on the molecular centroid to achieve learning, updating, and denoising of the different atomic and chemical bond features within the drug molecule. Finally, at the molecular feature level, we further leverage the attention mechanism, and through supervised learning, contrastive learning, and structured learning, we uncover the potential interaction relationships between different drug molecules, effectively detecting and defending against potential network attacks, thereby predicting the interaction between any two drug molecules i and j . The overall framework of the drug interaction prediction model we proposed is shown in Fig. 1 and consists of five main modules.

A. Message Passing Atomic Feature Network Based on Attention Mechanism

The attention mechanism-based message passing atomic feature network includes (a) Molecular Center of Mass 3D Position Encoding Module and (b) Attention mechanism and elastic message passing atomic feature network module. A drug molecule is composed of different atoms and the chemical bonds associated between them. Given a drug molecule with n atoms, this molecule can be represented as a directed graph $G = (X, E)$, where X and E are the initial features of atoms and chemical bonds, respectively, obtained through the RDKit [19] cheminformatics library. X consists of n atoms within the drug molecule, where $x_i \in X \in R^{n \times f_n}$ represents the feature embedding information of atom node i , including information such as the atom's hybridization, formal charge, and connectivity, acquired through RDKit. E represents the chemical bonds associated between atoms, including features such as the type of chemical bond, whether it is an aromatic bond, whether it forms a ring, etc. $e_{uv} \in E \in R^{n \times n \times f_e}$ represents the edge feature embedding information from atom node i to node j . f_n and f_e represent the feature dimensions of nodes and edges, respectively. The topological adjacency matrix $ADJ \in R^{n \times n}$ is composed of the shortest paths between two atoms. We use the (a) Molecular Center of Mass 3D Position Encoding Module to calculate the mean of the atomic 3D coordinates in the drug molecule to determine the molecular center of mass coordinates, and then apply word embedding techniques to obtain atomic embedding information. The primary goal of this module is to effectively encode the three-dimensional structure

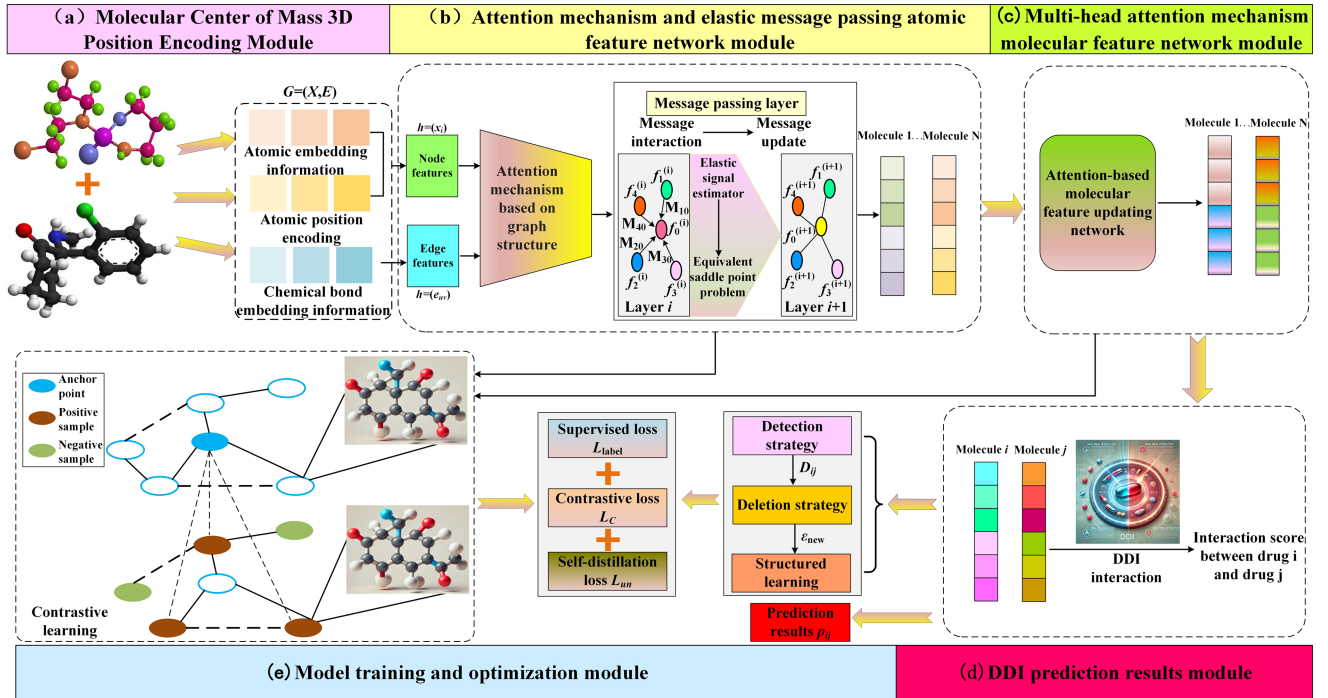


Fig. 1. Overall Architecture of the A3DPE-EMPGNN DDI Model. (a) The 3D geometric structure of the drug is introduced, and the centroid coordinates of the molecule are obtained by calculating the mean of the atomic 3D coordinates within the molecule, after which atomic encoding information is obtained through word embedding. (b) At the atomic level, a elastic message passing graph neural network and attention mechanism are employed to overcome the shortcomings of the inability to adaptively adjust the weight of information transmission between nodes and noise interference during message passing. (c) At the molecular level, a multi-head attention mechanism is used to learn interaction information between different drug molecules, and the learned drug molecule relationships are used to update the vector representation of each drug molecule. (d) The feature learning results of different drug molecules are fed into a multi-layer perceptron network and sigmoid function to obtain the DDI prediction results. (e) Detection and defense strategies against network attacks, combined with a binary cross-entropy loss function, self-distillation-based regularization, and unsupervised loss based on contrastive learning, are used to optimize the model while improving its robustness.

of atoms and the composition of chemical bonds, providing precise node and edge features for subsequent message passing and feature updating. This module provides atomic-level geometric information for DDI prediction, helping the subsequent model better understand the internal structure of the molecule.

In order to introduce the 3D geometric structure of the drug, we set the raw features of each atom as $\mathbf{a} \in R^{1 \times 3 \times n}$, represented by a three-dimensional matrix, where R denotes the set of real numbers, and n represents the vector dimension, which is a hyperparameter that determines the final dimension of the drug feature vector. The node feature matrix of the drug can be expressed as $\mathbf{A}^{(k)} \in R^{N \times 3 \times n}$, where N represents the total number of atom nodes. Thus, the atoms in the drug can be initialized as:

$$\alpha = (x, y, z) \cdot \mathbf{a}, \quad \alpha \in R^{1 \times 3 \times n} \quad (1)$$

where (x, y, z) represents the three-dimensional coordinates of the atom, and denotes the element-wise product between the three-dimensional coordinates and the three vectors in the atom's feature, generating a threedimensional representation of the same size.

We obtained the coordinates of the molecular centroid s_0 by calculating the mean v of the threedimensional coordinates of the n atoms in the molecule $v_i = (i = 1, 2, 3, \dots, n)$. The

Euclidean distance d_i between each atom and the molecular centroid was then calculated. The n atoms were then sorted by their distance d_i from the molecular centroid s_0 , from nearest to farthest, resulting in an atom index ordered by proximity to the centroid. This atomic index order was used as the positional encoding for the n atoms, and through word embedding, the positional encoding pos_i for the n atoms was obtained. The representations of the atomic feature $h(\mathbf{x}_i)$ and edge feature $h(\mathbf{e}_{uv})$, based on the 3D positional encoding of the molecular centroid, are as follows:

$$h(\mathbf{x}_i) = \mathbf{x}_i + \text{pos}_i \quad (2)$$

$$h(\mathbf{e}_{uv}) = \mathbf{e}_{uv} + h(\mathbf{x}_u) \quad (3)$$

Our 3D positional encoding method utilizes the spatial information of the molecular graph structure to describe the positional relationships between different atoms, overcoming the limitations of the molecular SMILES representation, which only uses the character input order in the SMILES sequence as positional encoding. This provides more structural information for learning the feature representations of atoms.

To address the issue that traditional elastic graph neural networks cannot effectively distinguish the different roles of nodes and edges, we propose the Attention Mechanism and Elastic Message Passing Atomic Feature Network Module, based on the

concept of interactive graph elastic graph neural networks [35]. In this module, different atoms in the drug molecule and their associated chemical bonds (edges) are represented as a directed graph, distinguishing between incoming and outgoing edges. By leveraging the elastic graph neural network based on the Transformer attention mechanism and combining it with our proposed 3D position encoding strategy based on the molecular center of mass, we calculate the attention scores for interactions between different atoms and chemical bonds in the drug molecule. We then perform message passing based on the graph structure, iteratively learning and updating the features of nodes and edges to obtain the feature representation of the drug molecule. The goal of this module is to enhance the update of atomic features and the accurate propagation of information, providing support for molecular-level interactions and feature extraction. This module directly impacts the effective extraction of molecular features and the accuracy of interaction prediction, making it an essential component of the atomic feature network. By representing the drug molecule as a directed acyclic graph, information from nodes and edges is propagated only in a determined direction, avoiding the cyclic updates of node and edge information caused by loops in undirected graphs, thus improving the efficiency of learning and updating node and edge information. Furthermore, as shown in equations (2) and (3), by distinguishing between incoming and outgoing edges, different nodes and their associated edges will possess distinct information. Through attention-based graph structure information transmission and update learning, it becomes easier to capture the different interaction information of various atoms and their associated edges, thereby extracting more effective drug molecule feature representations.

In the basic elastic graph neural network, the adjacency matrix is typically used for message aggregation and updating [36]. This approach has the drawback of considering only node features while ignoring edge information. Additionally, it cannot adaptively adjust the weights of information transmission between nodes during message passing. To address these issues, we introduced the attention mechanism. The method for calculating $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ is as follows:

$$\begin{cases} [\mathbf{Q}, \mathbf{V}] = h(\mathbf{X}) [\mathbf{W}^{\mathbf{Q}}, \mathbf{W}^{\mathbf{V}}] \\ \mathbf{K} = h(\mathbf{E}) \mathbf{W}^{\mathbf{K}} \end{cases} \quad (4)$$

where $\mathbf{Q}, \mathbf{V} \in R^{n \times d}$, $\mathbf{K} \in R^{n \times n \times d}$, $\mathbf{W}^{\mathbf{Q}}, \mathbf{W}^{\mathbf{V}} \in R^{f_n \times d}$, $\mathbf{W}^{\mathbf{K}} \in R^{f_e \times d}$ are learnable weight parameters.

To further explore the graph structure information, we treat each molecule as a directed graph and divide the edges into two types: incoming edges and outgoing edges. Using the $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ matrices obtained from equation (4), we calculate the corresponding message passing score matrices $\mathbf{M}_i$ and $\mathbf{M}_o$, respectively, as follows:

$$\begin{cases} \mathbf{M}_i = \text{einsum}(\mathbf{Q}_i, \mathbf{K}_i) \\ \mathbf{M}_o = \text{einsum}(\mathbf{Q}_o, \mathbf{K}_o) \end{cases} \quad (5)$$

where $\mathbf{M}_i, \mathbf{M}_o \in R^{n \times n}$, and einsum refers to Einstein summation convention. $\mathbf{Q}_i, \mathbf{K}_i, \mathbf{Q}_o, \mathbf{K}_o$ represent the corresponding matrices for incoming and outgoing edges, respectively. The

final message passing score matrix $\mathbf{M}$ is:

$$\mathbf{M} = \text{softmax}(\mathbf{M}_o + \mathbf{M}_i - \text{diag}\{\mathbf{M}_o\}) \quad (6)$$

where $\mathbf{M} \in R^{n \times n}$, and softmax normalizes the numerical vector into a probability distribution. $\text{diag}\{\mathbf{M}_o\}$ represents retaining only the diagonal elements of matrix $\mathbf{M}_o$.

To effectively address the noise issue in non-smooth signals and better capture the latent relationships between drug atom features, we propose a structured learning method based on the elastic graph neural network. This method utilizes the elastic graph neural network combined with the ℓ_{21} -norm to transform the denoising problem into an equivalent saddle point problem. The elastic graph neural network not only handles each feature dimension independently but also considers the correlations between feature dimensions.

The expression for the elastic graph signal estimator based on the ℓ_{21} -norm is:

$$\arg \min_{\mathbf{F} \in R^{n \times d}} \underbrace{\lambda_1 \|\tilde{\Delta} \mathbf{F}\|_{21}}_{g_{21}(\tilde{\Delta} \mathbf{F})} + \underbrace{\frac{\lambda_2}{2} \text{tr}(\mathbf{F}^T \tilde{\mathbf{L}} \mathbf{F}) + \frac{1}{2} \|\mathbf{F} - \mathbf{M}\|_{\mathbf{F}}^2}_{f(\tilde{\mathbf{F}})} \quad (7)$$

where $\mathbf{M} \in R^{n \times d}$ is the d -dimensional input signal of n nodes.

The first term $\|\tilde{\Delta} \mathbf{F}\|_{21}$ can be expressed in an edge-centric form as:

$$\|\tilde{\Delta} \mathbf{F}\|_{21} = \sum_{(v_i, v_j) \in \mathcal{E}} \left\| \frac{\mathbf{F}_i}{\sqrt{d_i + 1}} - \frac{\mathbf{F}_j}{\sqrt{d_j + 1}} \right\|_2 \quad (8)$$

Equation (8) is a penalty term based on the ℓ_{21} -norm, which penalizes the absolute differences between connected nodes in the graph. This increases the row sparsity of $\tilde{\Delta} \mathbf{F}$ and enforces similar sparse patterns across feature dimensions, thereby helping the model better capture the latent relationships between nodes.

The second term $\text{tr}(\mathbf{F}^T \tilde{\mathbf{L}} \mathbf{F})$ represents penalizing differences through a quadratic form, and its specific expression is:

$$\text{tr}(\mathbf{F}^T \tilde{\mathbf{L}} \mathbf{F}) = \sum_{(v_i, v_j) \in \mathcal{E}} \left\| \frac{\mathbf{F}_i}{\sqrt{d_i + 1}} - \frac{\mathbf{F}_j}{\sqrt{d_j + 1}} \right\|_2^2 \quad (9)$$

The last term $\|\mathbf{F} - \mathbf{M}\|_{\mathbf{F}}^2$ represents fidelity, ensuring the similarity between the result and the input signal.

Elastic Message Passing (EMP) is an efficient strategy designed to handle non-smooth graph signals. For typical ℓ_2 -norm-based graph smoothers, the message passing scheme can be computed using gradient descent iterations. However, in DDI prediction, when faced with various types of noise interference, the elastic graph signal estimator alone cannot meet the denoising requirements. These noise sources include measurement errors related to enzymes, targets, and chemical structures, as well as erroneous edge connections and unreasonable edge weights generated when constructing the graph structure. Moreover, the tight coupling between the two components $f(\mathbf{F})$ and $g(\tilde{\Delta} \mathbf{F})$, caused by the graph differential operator $\tilde{\Delta}$, is inseparable. We employ the EMP method to transform the original problem into an equivalent saddle point problem, as follows:

$$\min_{\mathbf{F}} \max_{\mathbf{Z}} f(\mathbf{F}) + \langle \tilde{\Delta} \mathbf{F}, \mathbf{Z} \rangle - g^*(\mathbf{Z}) \quad (10)$$

A three-step iterative algorithm is used to solve this saddle point problem.

First, the prediction step is performed, where the updated value of $\mathbf{F}$ is predicted by applying gradient descent to the original variables:

$$\bar{\mathbf{F}}^{k+1} = \mathbf{F}^k - \gamma \nabla f(\mathbf{F}^k) - \gamma \tilde{\Delta}^\top \mathbf{Z}^k \quad (11)$$

Next, a proximal dual ascent iteration is performed, which is an iteration on the dual variables:

$$\mathbf{Z}^{k+1} = \text{prox}_{\beta g^*}(\mathbf{Z}^k + \beta \tilde{\Delta} \bar{\mathbf{F}}^{k+1}) \quad (12)$$

Finally, an update is performed through another gradient descent step to produce the next iteration of $\mathbf{F}$:

$$\mathbf{F}^{k+1} = \mathbf{F}^k - \gamma \nabla f(\mathbf{F}^k) - \gamma \tilde{\Delta}^\top \mathbf{Z}^{k+1} \quad (13)$$

To handle the non-smoothness of the original problem, EMP employs a specific proximal operator. For the ℓ_{21} -norm, the proximal operator expression is:

$$\left(\text{prox}_{\beta g_{21}^*}(\mathbf{X})\right)_i = \frac{\mathbf{x}_i}{\|\mathbf{x}_i\|_2} \min(\|\mathbf{x}_i\|_2, \lambda_1) \quad (14)$$

The EMP method is an advanced strategy that combines saddle point problems and proximal operator techniques. The use of EMPGNN is aimed at enhancing the model's robustness and accuracy in DDI prediction tasks. In particular, it can effectively reduce noise and interference when dealing with complex drug molecular graph data, improving the model's performance on incomplete or attacked graph data. Its core advantage lies in the flexible node information transmission weight adjustment mechanism, which enables it to handle varying data conditions in practical applications. Especially when processing complex 3D structural information and interaction patterns between drug molecules, it can better adapt to the DDI prediction requirements of different scenarios.

After a single layer of message passing, the update equations for the node and edge features are as follows:

$$\begin{cases} h(\mathbf{X}) = \text{matmul}(\mathbf{F}, \mathbf{V}) \\ h(\mathbf{E}) = \mathbf{F} \odot \mathbf{K} \end{cases} \quad (15)$$

where matmul represents matrix multiplication, and represents element-wise multiplication between two matrices. We use average pooling to generate the molecular representation from the output of the atomic feature network, as follows:

$$\mathbf{F}\mathbf{a}_i = \frac{1}{K_i} \sum_{k=1}^{K_i} h(\mathbf{X}^m)_k \quad (16)$$

where $\mathbf{F}\mathbf{a}$ is the matrix composed of all molecular features, $\mathbf{F}\mathbf{a}_i$ represents the feature representation of the i -th molecule, K_i represents the maximum number of atoms in the i -th molecule, and $h(\mathbf{X}^m)_k$ represents the feature of the k -th atom in the m -th molecule after being updated by the aforementioned network.

The attention mechanism-based message passing atomic feature network ensures the precise extraction and effective propagation of features from atomic structures to molecular features,

providing reliable feature inputs for the subsequent DDI prediction module, thereby enhancing the overall model's prediction accuracy and robustness.

B. Molecular Feature Network Based on Multi-Head Attention Mechanism

The atomic feature network primarily focuses on the internal structural information of individual molecules, describing the geometry and chemical properties of the molecule by extracting and encoding atomic features. However, such a network is limited to the internal representation of the molecule and overlooks the interaction information between different drug molecules. If DDI prediction is directly based on the molecular features output by the atomic feature network, it will not effectively capture the interaction relationships between drug molecules, potentially leading to a decrease in prediction accuracy. To overcome this issue, we designed a molecular feature network based on a multi-head attention mechanism, which includes the (c) Multi-head Attention Mechanism Molecular Feature Network Module and the (d) DDI Prediction Results Module. The molecular representations output by the atomic feature network are passed as inputs to the molecular feature network. In this process, we introduce the multi-head attention mechanism to learn the interaction information between different drug molecules. The multi-head attention module can automatically adjust the weight of each drug molecule's features by focusing on the relationships between different drug molecules, and update the vector representation of each drug molecule. By utilizing the updated drug molecule features, we can perform DDI prediction more accurately, thus improving the precision and reliability of the results. This process, by incorporating cross-molecular interaction learning, enables the model to comprehensively consider the relationships between drug molecules, thereby optimizing the effectiveness of DDI prediction.

By applying a linear transformation to the molecular feature representation $\mathbf{F}\mathbf{a}$ output by the atomic feature network, we obtain the matrices $\mathbf{Q}'$, $\mathbf{K}'$, $\mathbf{V}'$, and calculate the multi-head attention as follows:

$$\text{head} = \text{softmax}\left(\frac{\mathbf{Q}'\mathbf{K}'^T}{\sqrt{d_k}}\right) \mathbf{V}' \quad (17)$$

$$\text{MultiHead} = \text{concat}(\text{head}_1, \text{head}_2, \text{head}_3) \quad (18)$$

where $\mathbf{Q}' \in R^{n' \times d'}$, $\mathbf{K}' \in R^{n' \times d'}$, $\mathbf{V}' \in R^{n' \times d'}$, n' represents the number of molecules, and concat refers to the concatenation operation.

To prevent the vanishing gradient problem caused by too many network layers and to achieve the fusion of features at different levels, we use a dense connection structure between the linear layers in the Multi-Layer Perceptron (MLP) network to compute the final output: the drug molecular feature representation $\mathbf{F}\mathbf{m}$. As follows:

$$\mathbf{x}_l = \text{layer}^l\left(\sum_{k=0}^{l-1} \mathbf{x}_k\right) \quad (19)$$

$$\mathbf{Fm} = \text{DenseMLP}_N(\mathbf{x}_0) = \text{layer}^N \left(\sum_{k=0}^{N-1} \mathbf{x}_k \right) \quad (20)$$

where $\mathbf{x}_0$ is the output of the multi-head attention layer, $\mathbf{x}_l$ represents the calculation formula for each densely connected layer, layer^l denotes the l -th linear layer, and DenseMLP_N represents an MLP network with N densely connected layers.

$$\mathbf{S}_{ij} = \mathbf{E}_i \odot \mathbf{E}_j \quad (21)$$

where $\mathbf{E} = \mathbf{Fm}$.

Given drug molecule pair i and j , the value S_{ij} is calculated using equation (21), and then passed through an MLP and a sigmoid function to obtain the final link prediction result p_{ij} .

C. Model Training and Optimization

In the DDI prediction process, network attacks may originate from malicious interference or external manipulation, affecting the model's normal operation and prediction results. Since DDI prediction relies on complex structured data processing methods such as GNNs, attackers can disrupt the network's structure by tampering with node features or the connections between edges, which in turn affects the propagation and learning of interaction information between molecules. For example, attackers may interfere with the model's input by adding fake nodes or edges, deleting critical nodes or edges, or altering node features, leading to distorted or erroneous prediction results. Moreover, due to the complexity of interactions between drug molecules, attackers may exploit this complexity by introducing noise during the training process, altering the true relationships between molecules, and further reducing the model's prediction accuracy. Therefore, to address the potential network attacks encountered during DDI prediction, we propose the (e) Model Training and Optimization Module. In this module, we have designed a detection defense strategy to enhance the model's stability and prediction accuracy when facing attacks. The core goal of this defense strategy is to minimize the negative impact of network attacks on molecular feature extraction, message passing, and prediction results. We have divided the entire defense strategy into three key steps.

Step 1: Detection strategy. To solve the elastic signal estimation problem defined in equation (7), the elastic message passing framework is used to learn the node representations of the attacked graph, enabling the calculation of signal differences across the edges in the graph. As follows:

$$D_{ij} = \left\| \frac{\mathbf{F}_i}{\sqrt{d_i + 1}} - \frac{\mathbf{F}_j}{\sqrt{d_j + 1}} \right\|_2 \quad (22)$$

where D_{ij} represents the signal difference between nodes $\mathbf{x}_i$ and $\mathbf{x}_j$. This difference indicates the signal gap between the two nodes, where a larger difference suggests that the edge between these two nodes is under attack or there is some kind of anomaly.

Step 2: Deletion strategy. To filter out potentially attacked edges from the graph, we introduce a threshold τ . All edges with a signal difference greater than τ will be considered anomalous

and removed from the graph.

$$\mathcal{E}_{\text{new}} = \mathcal{E} - \{(\mathbf{x}_i, \mathbf{x}_j) \mid D_{ij} > \tau\} \quad (23)$$

The purpose of this strategy is to ensure that our network only considers edges with stable signals that are likely not under attack.

Step 3: Structured learning. Simply removing potentially attacked edges is not enough. To further optimize the performance and robustness of the elastic graph neural network, we need to perform structured learning on the graph structure. Through this process, the elastic graph neural network will update its weights based on the new graph structure $\mathcal{E}_{\text{new}}$. This ensures that the elastic graph neural network can better capture the true feature relationships in the graph while eliminating the interference from attacked edges. Defining the network weights as $\mathbf{W}$, the feature propagation process under the original graph structure can be expressed as:

$$\mathbf{H}^{(l+1)} = \sigma(\mathbf{A}\mathbf{H}^{(l)}\mathbf{W}^{(l)}) \quad (24)$$

where $\mathbf{A}$ is the adjacency matrix of the graph, $\mathbf{H}^{(l)}$ represents the node features at layer l , and $\sigma(\cdot)$ is the activation function. Under the new graph structure $\mathcal{E}_{\text{new}}$, the feature propagation is updated as:

$$\mathbf{H}_{\text{new}}^{(l+1)} = \sigma(\mathbf{A}_{\text{new}}\mathbf{H}_{\text{new}}^{(l)}\mathbf{W}_{\text{new}}^{(l)}) \quad (25)$$

where $\mathbf{A}_{\text{new}}$ is the adjacency matrix of the new graph. Our goal is to optimize the new weights $\mathbf{W}_{\text{new}}$ through the structured learning process so that the network can better capture the true feature relationships under the new graph structure.

$$\mathcal{L}(\mathbf{W}_{\text{new}}) = \sum_{i \in \mathcal{V}} L(f(\mathbf{x}_i; \mathbf{W}_{\text{new}}), \mathbf{y}_i) \quad (26)$$

where $f(\mathbf{x}_i; \mathbf{W}_{\text{new}})$ represents the predicted value of node i under the new weights, while $\mathbf{y}_i$ is its true label, and L is the loss function. The weights $\mathbf{W}_{\text{new}}$ are updated using the gradient descent algorithm, reducing the interference caused by attacked edges, thereby enhancing the robustness of the model.

To improve the model's generalization ability, we introduce two types of unsupervised loss functions on top of the traditional Binary cross entropy (BCE) loss function L_{label} , including a self-distillation-based regularization constraint L_{un} and a contrastive learning-based unsupervised loss L_c . The overall loss function L of the model is as follows:

$$L = L_{\text{label}} + \alpha L_c + \beta L_{\text{un}} \quad (27)$$

where α and β are the weight coefficients for the corresponding loss functions. L_{label} represents the error loss of the model measured using the binary cross-entropy loss function, as follows:

$$L_{\text{label}} = \sum_{(i,j) \in Tr} (\text{BCE}(\mathbf{r}_{ij}, \mathbf{y}_{ij}) + \text{BCE}(\mathbf{p}_{ij}, \mathbf{y}_{ij})) \quad (28)$$

where Tr represents the set of sample pairs in the training set, $\mathbf{y}_{ij}$ denotes the true label of the sample pair (i, j) and $\mathbf{r}_{ij}$ and $\mathbf{p}_{ij}$ represent the predicted results output by the atomic feature network and the molecular feature network, respectively.

The self-distillation-based regularization constraint [37] uses the output of the molecular feature network to distill the output of the atomic feature network, further improving the quality of the features produced by the atomic feature network. We use Kullback-Leibler divergence (KL divergence) to represent L_{un} , as follows:

$$L_{un} = \sum_{(i,j) \in D \setminus Tr} \text{KL}(\mathbf{p}_{ij} \parallel \mathbf{r}_{ij}) \quad (29)$$

where $\text{KL}(\mathbf{p}_{ij} \parallel \mathbf{r}_{ij})$ represents the KL divergence between the two probability distributions, used to measure the difference between them. $D \setminus Tr$ denotes the set of sample pairs outside the training set.

For the contrastive learning-based unsupervised loss function, the output feature of each drug molecule from the molecular feature network is selected as the anchor, while its first-order neighbors and non-first-order neighbors in the atomic feature network's output are treated as positive and negative samples, respectively. Through contrastive loss learning, the anchor is brought closer to its positive samples and distinguished from its negative samples. As follows:

$$(\hat{\omega}, \hat{\Phi}, \hat{\varphi}) = \arg \max_{\omega, \Phi, \varphi} \sum_{i \in D} \frac{1}{\tilde{C}(i)} \times \sum_{j \in C(i) \cap \{i\}} \hat{I}_{\omega, \Phi, \varphi}(\mathbf{F}\mathbf{a}_i^{\Phi}; \mathbf{F}\mathbf{m}_j^{\varphi}) \quad (30)$$

where $C(i)$ and $\tilde{C}(i)$ represent the set of first-order neighbors and non-first-order neighbors of node i , respectively. Φ and φ define the parameters of the atomic feature network and molecular feature network, respectively, and $\hat{I}_{\omega, \Phi, \varphi}$ defines the mutual information estimator.

Since mutual information cannot be directly optimized, we use Jensen-Shannon Divergence (JSD) to optimize the lower bound of mutual information [38]. By minimizing the contrastive loss function L_c , mutual information is maximized. The formula for the contrastive loss function is as follows:

$$L_c = \frac{1}{|Tr|} \sum_{i \in Tr} \left(\frac{1}{|C(i)|} \sum_{j \in C(i)} \text{JSD}(\mathbf{F}\mathbf{m}_i, \mathbf{F}\mathbf{m}_j) - \frac{1}{|\tilde{C}(i)|} \sum_{k \in \tilde{C}(i)} \text{JSD}(\mathbf{F}\mathbf{m}_i, \mathbf{F}\mathbf{m}_k) \right) \quad (31)$$

IV. EXPERIMENT AND ANALYSIS

A. Dataset and Evaluation Metrics

To validate the effectiveness and superiority of our method, we conducted experiments using two real-world datasets. We selected two commonly used DDI datasets, ZhangDDI [39] and ChCh-Miner [40], to conduct comparison experiments, ablation experiments, and parameter experiments to verify the effectiveness and superiority of the method proposed in this paper. ZhangDDI contains 548 drugs and 48 548 drug interaction pairs, while ChCh-Miner contains 1514 drugs and 48 514 drug interaction pairs. In this study, these two datasets are referred

to as ZhangDDI and ChCh-Miner, respectively. We used four different metrics to evaluate our model, including ACC (Accuracy), AUC (Area Under the Curve), AP (Average Precision), and F1-score.

ACC: Defined as the number of correct predictions divided by the total number of predictions. AUC: Represents the probability that a classifier ranks a randomly chosen positive instance higher than a randomly chosen negative instance. AP: Obtained by taking the mean of the average precision across all classes. F1-score: The harmonic mean of precision and recall.

B. Baseline Experiment and Analysis

We compared our method with eight existing DDI prediction methods. These eight baseline methods are currently the most advanced graph-structure-based deep learning methods for DDI prediction. These methods are further divided into two categories: DDI prediction methods based on molecular feature networks and those based on substructure features. During the experiments, the parameter settings of the baseline methods were kept mostly consistent with those in the original papers. When issues such as slow gradient descent, long convergence times, or difficulty in model convergence occurred, parameters such as the learning rate and batch size were appropriately adjusted.

For the experiments on the ZhangDDI and ChCh-Miner datasets, we followed the data splitting method described in reference [41] to ensure consistency with existing studies. Specifically, all data samples were split in a 4:1 ratio, with 80% of the samples used for the training set and 20% for the test set. Within the training set, we further randomly selected 1/4 of the samples as the validation set to allow for model tuning and early stopping during training. To prevent overfitting and ensure the reliability of the experimental results, we set a stopping criterion such that training would halt after 10 consecutive epochs without improvement in the best validation accuracy, ensuring the efficiency of model training. All experimental results were obtained by conducting five independent experiments and performing statistical analysis, which helps to effectively minimize the impact of random errors on the results and improve the stability and credibility of the findings. For the experiments, we set the dimensionality of atomic features to 120 and chemical bond features to 20 in the atomic feature network. This configuration effectively captures the atomic and chemical bond information within the molecule, providing sufficient feature support for subsequent model training. In the molecular feature network, we set the number of attention heads to 3 to extract diverse interaction information between molecules, enhancing the model's representational power. The coefficients in the objective function were set to 1.0 and 0.8, respectively, to balance the weights between different loss terms, facilitating better convergence during the optimization process. To highlight the experimental results, the same model parameter settings were used for both datasets. No specific adjustments were made during the experiment, and all parameters, such as learning rate, batch size, etc., were kept consistent across both datasets. All experiments were conducted on the Pytorch 1.6.0 platform. These settings not only

TABLE I
COMPARISON OF EXPERIMENTAL RESULTS BETWEEN THE A3DPE-EMPGNN MODEL AND BASELINE METHODS

Datasets	Evaluation metrics	GraphDDI	MTrans	RaGSECo	RCAN-DDI	EmerGNN	GPF	MRCGNN	DDI-GCN	Ours
ZhangDDI	ACC	94.27	96.37	94.12	96.12	96.32	97.21	95.47	94.25	98.61
	AUC	95.63	94.42	94.13	95.85	97.13	94.51	95.13	96.26	98.35
	AP	94.07	96.74	94.94	93.49	93.08	96.42	95.54	93.17	99.24
	F1	96.02	94.38	97.18	95.21	96.11	94.72	95.25	96.92	99.16
ChCh-Miner	ACC	95.41	96.02	93.67	95.96	94.61	96.13	96.02	95.58	98.27
	AUC	94.74	93.68	95.05	96.06	96.84	95.39	94.69	95.83	98.62
	AP	93.82	95.75	95.35	92.85	94.51	95.58	96.17	94.18	98.96
	F1	95.76	93.21	96.59	95.82	95.81	94.51	96.73	97.13	99.04

ensure the standardization of the experiment but also provide a clear reference framework for future model optimization and improvements.

GraphDDI [42] combines interaction graphs and drug features to create a unified representation of drug molecules. The model connects these representations and employs a feedforward neural network to predict DDI.

MTrans [43] is a deep learning model designed to capture structural and pharmacological drug features across multiple semantic spaces, enhancing DDI prediction through multi-head attention and knowledge graph integration.

RaGSECo [44] uses relation-aware graph embeddings and co-contrastive learning for improved drug-drug interaction prediction by integrating multi-relational and multi-attribute graphs.

RCAN-DDI [45] combines a relationship-aware structural feature learning module with a topological feature learning module based on the DDI network to capture multimodal features of drugs.

EmerGNN [46] leverages biomedical network information to predict interactions for emerging drugs, using a flow-based GNN approach.

GPF [47] discusses a universal prompt-based tuning method called Graph Prompt Feature for pre-trained GNN models, adapting pre-trained models for DDI prediction tasks.

MRCGNN [48] integrates the two aspects of information by deploying a GNN on the multi-relational DDI event graph attributed with the drug features extracted from drug molecular graphs.

DDI-GCN [49] uses GCNs combined with co-attention mechanisms to predict DDIs and highlight relevant substructures within drug molecules. The model is trained on DDI binary and categorical classification tasks.

As observed in Table I, compared to the eight baseline methods, our method improved the ACC, AUC, AP, and F1 metrics on the ZhangDDI to 98.61%, 98.35%, 99.24%, and 99.16%, respectively. On the ChCh-Miner, the corresponding results improved to 98.2%, 98.62%, 98.96%, and 99.04%. These four metrics outperform the current state-of-the-art methods. This is because DDI prediction methods based on molecular feature networks learn the interaction relationships between different drug molecules directly at the molecular level, ignoring the internal structural characteristics of the molecules. As a result, the performance of these algorithms is limited by the quality of the molecular feature representation. In contrast, our method first learns the features of each drug molecule at the atomic level, then combines supervised learning and contrastive learning to

continuously optimize the learned molecular features. Based on these molecular features, the method learns the interaction relationships between different drug molecules, providing a theoretical advantage. The experimental results also demonstrate the superiority of our proposed method.

Compared to substructure-based DDI prediction methods, although these methods can extract drug molecule features from both the atomic and molecular levels, they lack a clear mechanism to learn the different attention information between atoms and chemical bonds, as well as between molecules. In contrast, our method simultaneously considers message passing between different atoms and their associated edges, and it can learn attention information between different atoms. Our approach enables attention learning at both the atomic and molecular levels, with each level serving distinct functions. Additionally, to address the issues of noise and vulnerability to network attacks faced by existing graph neural networks, which result in insufficient model robustness, we propose a structured learning method based on a general elastic message passing scheme. By utilizing the ℓ_{21} -norm, this method can efficiently denoise graph signals affected by noise. The elastic message passing strategy effectively addresses the non-smoothness issues related to graph structures. In the presence of adversarial graph structures, our designed detection and defense strategy enhances the robustness of the network. Therefore, our method demonstrates superior overall performance.

C. Ablation Experiment and Analysis

To validate the effectiveness of the proposed model, we conducted ablation experiments on both datasets, focusing on the multi-layer attention mechanism, 3D position encoding, anti-interference mechanism, and the loss function. Our ablation experiments tested the model's performance from different perspectives, and the experimental results are shown in Figs. 2 and 3. In these figures, A3DPE-EMPGNN represents the experimental results without any ablation. Fig. 2 shows the ablation experiment results on ZhangDDI, where the size of the sector area indicates the maximum result across different training batches. Fig. 3 presents the ablation experiment results on ChCh-Miner across different training batches, with the x-axis representing the training batches of the model.

In the ablation experiment on the multi-layer attention mechanism, to validate the effectiveness of the multi-level attention mechanism proposed in this paper, we conducted ablation experiments on both datasets for the attention mechanism-based

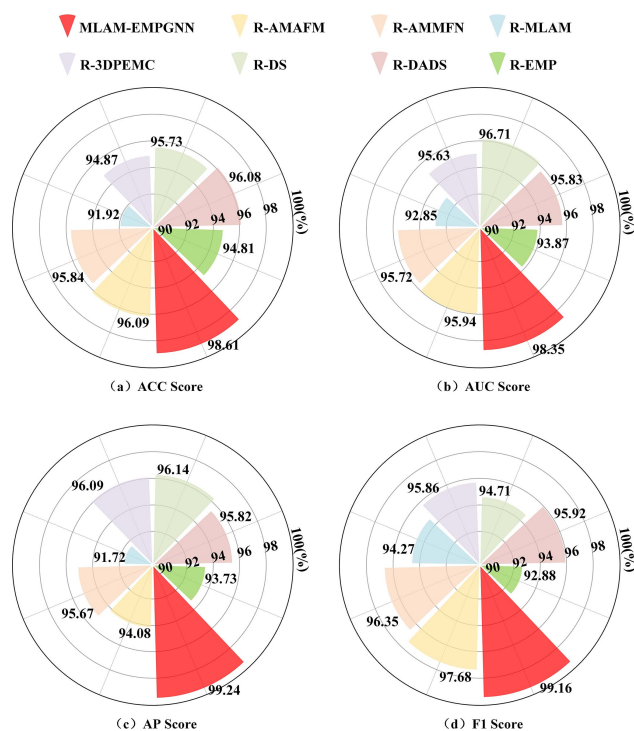


Fig. 2. Results of different ablation experiments on ZhangDDI. R-AMAFM represents the ablation experiment of the atomic feature network. R-AMMFN represents the ablation experiment of the molecular feature network. R-MLAM represents the ablation experiment of both the atomic and molecular feature networks. R-3DPEMC represents the ablation experiment of the 3D position encoding. R-DS represents the ablation experiment of the elastic graph signal estimator. R-DADS represents the ablation experiment of the detection defense strategy. R-EMP represents the ablation experiment of both the elastic graph signal estimator and the detection defense strategy.

atomic feature network and molecular feature network separately, as well as a joint ablation experiment for both the atomic and molecular feature networks.

The ablation experiment results for the atomic feature network and molecular feature network are shown in Figs. 2 and 3.

R-AMAFM demonstrates that by introducing an attention mechanism, the model can effectively capture important atomic information at the atomic level within the molecule. If this mechanism is removed, the model will not be able to precisely capture the interactions between atoms, or between atoms and chemical bonds. This will result in the loss of atomic feature information in the molecular representation, which subsequently affects the overall quality of the drug molecule features and the accuracy of interaction predictions.

R-AMMFN focuses on molecular-level feature learning, utilizing the attention mechanism to extract information about interactions between drug molecules. Without the molecular feature network, the model will fail to recognize interactions between drug molecules and can only independently predict the effects of each drug based on atomic features, losing crucial interaction information between molecules. As a result, the model's prediction ability will significantly decline, especially when dealing with drug pairs with complex interactions, leading to inaccurate predictions. Therefore, the attention mechanisms

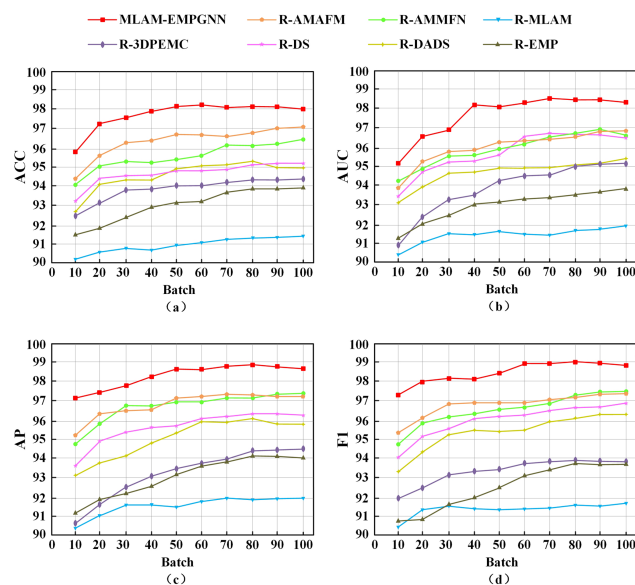


Fig. 3. Results of different ablation experiments on ChCh-Miner.

for both atomic and molecular features are key to improving DDI prediction accuracy.

R-MLAM focuses on multi-level feature fusion, where the atomic and molecular feature networks integrate atomic information within the drug molecule and interaction information between molecules at a higher level, thereby enhancing the model's comprehensiveness and accuracy. Experimental results show that when both the atomic and molecular feature networks are ablated, the model's performance drops significantly, especially in terms of evaluation metrics such as ACC, AUC, AP, and F1, which are noticeably lower than models based on the multi-level attention mechanism. This indicates that the combination of atomic and molecular feature networks is indispensable for drug interaction prediction.

R-AMAFM, R-AMMFN, and R-MLAM, by focusing separately or jointly on atomic-level and molecular-level features, greatly improve the quality of drug molecule features and the model's predictive ability. Without these modules, the model cannot fully understand the complex interactions between drug molecules, leading to a significant decrease in prediction accuracy. Therefore, the multi-level attention mechanism-based feature network approach we propose effectively enhances model performance, providing important support for drug interaction prediction.

In the ablation experiment of 3D position encoding, the experimental results are shown in Figs. 2 and 3. From the results, it can be observed that when the 3D position encoding method is removed, the model's performance significantly decreases, and its stability also worsens. Without the 3D position encoding, the model relies solely on the order of atomic nodes in the SMILES sequence to perform atomic feature learning, while neglecting the spatial relationships between atoms within the molecule. The SMILES sequence is essentially a one-dimensional linear representation of a molecule, which only provides information about the sequence of atoms but cannot express the relative

positions of atoms in three-dimensional space. As a result, without the support of 3D spatial position information, the model cannot fully learn the spatial distribution and interaction patterns between atoms in the drug molecule, thus affecting the accurate representation of drug molecule features. When we use the molecular center of mass-based 3D position encoding method, the model is able to more accurately capture the spatial structural information of the drug molecule. By incorporating the center of mass and 3D coordinates, the model not only learns the position of each atom but also understands the geometric relationships between atoms within the molecule, thus improving the efficiency of atomic feature encoding. The molecular center of mass-based 3D position encoding method is indispensable for drug interaction prediction. It effectively enhances the model's accuracy and robustness, especially demonstrating significant advantages when handling complex drug molecule interactions.

In the ablation experiment of the anti-interference mechanism, to validate the effectiveness of the structured learning approach of our proposed elastic graph neural network, we performed the R-DS, R-DADS, and R-EMP ablation experiments on two datasets. The experimental results are shown in Figs. 2 and 3.

The R-DS ablation experiment primarily aims to remove noise from graph signals by applying the ℓ_{21} -norm. In drug interaction prediction tasks, graph data often contains noise, such as incorrect edge connections, missing edges, and inaccurate node features. This noise causes irregularities in the graph structure, which in turn affects the model's prediction accuracy and robustness. When the elastic graph signal estimator is removed, the model is unable to effectively remove this noise during the learning process, leading to a deterioration in the quality of graph structure information and a significant decline in the model's performance, especially when dealing with noisy data, which greatly reduces robustness. This indicates that the elastic graph signal estimator, through estimating graph signals and removing noise, significantly enhances the model's accuracy and stability in drug interaction prediction.

The R-DADS ablation experiment demonstrates that the detection defense strategy is a key module applied during the training optimization process, primarily used to defend against various potential attacks that the model might face. The model may encounter attacks such as adversarial attacks, data poisoning, and model stealing. These attacks not only reduce prediction accuracy but may also lead to misleading results or cause the model to misjudge the interactions between drugs. By introducing the detection defense strategy, the model can effectively recognize and defend against these potential attacks, maintaining high efficiency in predicting real data. When the detection defense strategy is removed, the model's robustness significantly decreases, and its prediction performance declines notably when faced with adversarial attacks. Therefore, the detection defense strategy not only improves the model's ability to identify abnormal disturbances but also enhances its safety and stability in practical applications.

The R-EMP ablation experiment showed the most significant performance drop. This demonstrates that the combination of the elastic graph signal estimator and the detection defense strategy

TABLE II
COMPARISON OF THE IMPACT OF LOSS FUNCTIONS ON
MODEL PERFORMANCE

Datasets	Algorithm	ACC	AUC	AP	F1
ZhangDDI	T1	96.43	95.57	96.18	91.34
	T2	94.15	93.36	95.33	89.16
	T3	95.42	92.38	93.96	92.15
	T4	92.57	93.94	94.56	95.28
	Ours	98.61	98.35	99.24	99.16
ChCh-Miner	T1	95.23	94.96	95.46	93.27
	T2	93.85	93.64	94.05	91.98
	T3	94.39	93.21	94.66	93.32
	T4	94.15	95.06	95.59	95.24
	Ours	98.27	98.62	98.96	99.04

provides the graph neural network with powerful defense capabilities, allowing it to maintain high prediction performance even in environments with substantial interference. Through the dual effect of the elastic graph signal estimator and the detection defense strategy, the model is able to perform deeper structured learning, which not only optimizes the structural representation capability of the graph but also enhances its resistance to noise and attacks. This significantly improves the overall performance and robustness of the model.

In the ablation experiments on the loss function, to validate the effectiveness of the self-distillation constraint term and the contrastive learning loss term we introduced, we conducted ablation experiments on both loss terms across two datasets. Furthermore, we replaced the contrastive learning loss term with the Mutual Information Noise Contrastive Estimation (infoNCE) model [22] to examine the impact of different contrastive loss functions on model performance. Additionally, we modified the sampling method for positive and negative samples to test its influence on the contrastive learning loss function used in this study. For each drug molecule, we selected its output features from the molecular feature network as an anchor. We chose two sampling methods for positive and negative samples. One method, which was used in our experiments, involves taking the output features of the anchor's first-order neighbors and non-first-order neighbors from the atomic feature network as positive and negative samples, respectively. The other method involves taking the output features of the anchor's second-order neighbors as positive samples and other nodes as negative samples for contrastive learning.

Table II presents the comparison of experimental results showing the impact of different loss functions on model performance. T1 indicates the removal of the self-distillation constraint term, T2 indicates the removal of the contrastive learning loss term, T3 refers to the infoNCE contrastive loss term, and T4 refers to the contrastive loss term with different sampling methods. It can be observed that, for both ZhangDDI and ChCh-Miner, the model performance decreases to some extent when either the self-distillation-based regularization constraint term or the contrastive learning-based loss term is removed, indicating that both loss functions are indispensable for improving model performance. Regarding the two sampling methods for positive and negative samples, the experimental results show that the sampling method used in this study performs better. This result also suggests that the sampling

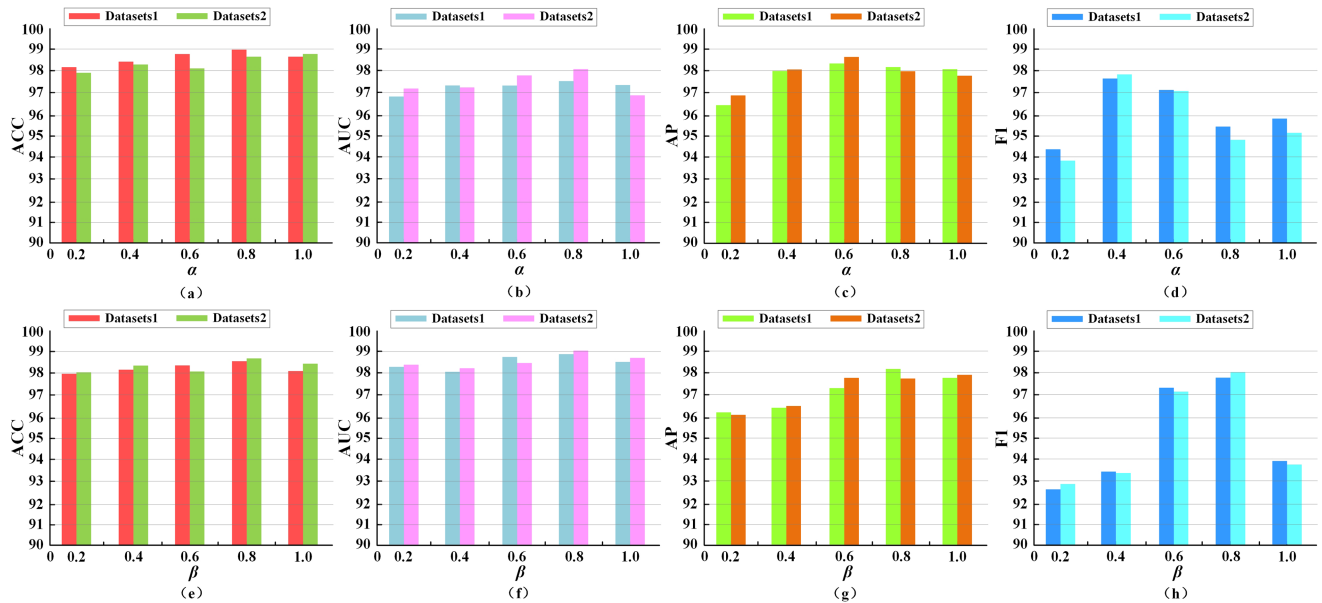


Fig. 4. Impact of different values of α and β on model performance in ZhangDDI and ChCh-Miner.

method for positive and negative samples has a certain impact on model performance. As for different contrastive learning loss functions, our method achieves the best results across both datasets and is relatively more stable. This is because infoNCE optimizes the lower bound of mutual information through self-normalized importance sampling, which requires a relatively larger number of negative samples. In contrast, the JSD method we used is less sensitive to the number of negative samples, resulting in more stable performance. Therefore, different contrastive learning loss functions have some impact on model performance, and the contrastive learning loss function used in this study has certain advantages.

D. Parameter Experiment and Analysis

Different parameters directly affect the performance of the model. We conducted a sensitivity analysis on the parameters α, β in the loss function, and the threshold τ in the detection defense strategy. To analyze the sensitivity of the three parameters, α was set to values from $\{0.2, 0.4, 0.6, 0.8, 1\}$, β to values from $\{0.2, 0.4, 0.6, 0.8, 1\}$, and τ to values from $\{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1\}$. When analyzing the sensitivity of α, β and τ were fixed at 0.8; when analyzing the sensitivity of β, α and τ were fixed at 1. When analyzing the sensitivity of τ, α and β are fixed, with values set to 0.6. At the beginning of the experiment, the initial values of 0.6, 0.8, and 1 were set based on randomly selected starting values. In the subsequent sensitivity analysis process, we gradually adjusted these parameters through the control variable method and grid search, and observed their impact on the model's performance. Fig. 4 shows the experimental results of parameters α and β on both datasets. It can be observed that, for both datasets, different values of α and β have relatively little effect on the results of ACC, AUC and AP for our method, while they have a relatively larger impact on the F1 score. From Figs. 5 and 6, it can be seen

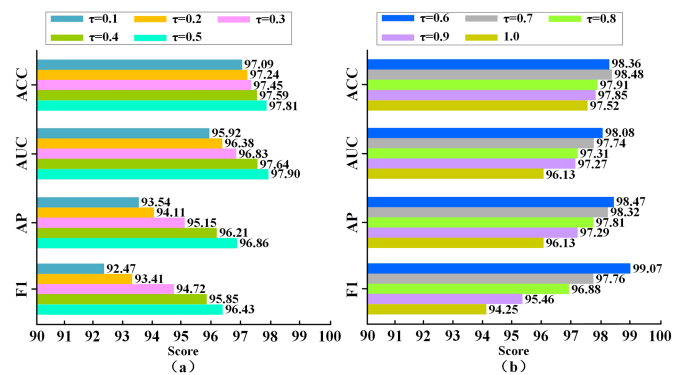


Fig. 5. Impact of different values of τ on model performance in ZhangDDI.

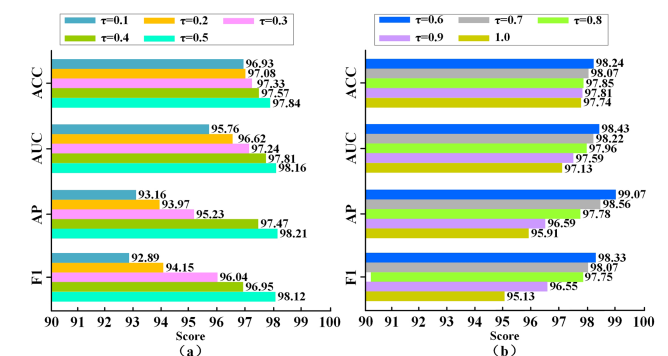


Fig. 6. Impact of different values of τ on model performance in ChCh-Miner.

that different values of τ have a greater impact on our method's performance in terms of AUC, ACC, AP, and F1. Through a comprehensive sensitivity analysis of the parameters α, β , and τ on the model's performance, we found that during the control variable method step-by-step selection of the optimal values

for α , β , and τ , the best experimental results were achieved when $\alpha=0.4$, $\beta=0.8$, and $\tau=0.6$. However, to ensure that this parameter combination is truly globally optimal and not just locally optimal, we further validated the result using grid search. During the grid search process, we refined the search range around $\alpha=0.4$, $\beta=0.8$, and $\tau=0.6$, tested neighboring parameter combinations, and compared their experimental performance. The final results confirmed that this parameter combination was indeed the best. By combining the control variable method with grid search, we ensured the reliability and reproducibility of the experimental conclusions. This optimal parameter combination search strategy combines the efficiency of the control variable method with the comprehensiveness of grid search, making parameter adjustment both precise and efficient.

V. CONCLUSION

To address the accuracy requirements for DDI prediction, we propose the A3DPE-EMPGNN DDI method, which aims to tackle the impact of atomic 3D structures and graph noise on DDI prediction results. We design an atomic feature and molecular feature network based on a multi-layer attention mechanism to learn feature information from both molecular internal and external levels, and improve molecular encoding efficiency by incorporating 3D position encoding based on molecular centroids. Additionally, we enhance the robustness of the graph neural network through an elastic message-passing method, and the advanced nature of the model has been validated through experiments. This method provides a more effective tool for DDI prediction and new drug development.

Our model still has some limitations. When processing complex or unknown drug molecules, it is constrained by the scale of the dataset and the incompleteness of drug information, leading to insufficient generalization capability. Although the model incorporates elastic graph neural networks and adversarial defense mechanisms, its robustness and stability are still challenged in certain specific scenarios or when facing extreme data, and it cannot completely avoid potential errors. Future research could integrate multimodal information such as drug interaction relationships from textual data and molecular graphs to further enhance the model's prediction capability. Furthermore, exploring more efficient graph neural network architectures and optimization algorithms could improve the model's stability when facing extreme data or potential network attacks and enhance its computational efficiency when handling large-scale molecular data.

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